

Table 6

Model components and regression variable results obtained from the best-fit (first-order polynomial model) model for single-walled carbon nanotubes (SWCNTs).

Independent and original variables	SE ^a	t-ratio	p-Value ^b
$Y_2 = \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \beta_4(X_4) + \beta_0 E(\text{SWCNTs}) = (-2.569)(pH) - (2.024)(C_0) + (25.727)(M) + (0.199)(t) + 41.795$			
Solution pH ($X_1 = pH$)	0.1872	-13.7275	0.00000
Initial fluoride concentration ($X_2 = C_0$: mg/L)	0.2381	-8.5018	0.00000
Adsorbent dose ($X_3 = M$: g/L)	2.3412	10.9888	0.00000
Contact time ($X_4 = t$: min)	0.0115	17.4135	0.00000
$\beta_0 = \text{constant term}$	1.8023	23.1906	0.00000

^a Standard error.^b p-Values < 0.05 were considered to be significant.

the results in Table 2 that the theoretical value of the adsorption capacity is higher for MWCNTs than SWCNTs, and this is probably due to the higher porosity of MWCNTs. The maximum adsorption capacities of MWCNTs and SWCNTs were found to be 2.83 and 2.4 mg/g, respectively. It was obvious that the MWCNT adsorbent was more effective in removing fluoride from aqueous solutions. This could be due to the complex formation with MWCNTs as referenced earlier but a more likely explanation is the higher availability of internally available adsorption sites on MWCNTs due to the different pore size diameters resulting in a larger internal available surface area than for SWCNTs [43]. The adsorption of fluoride onto carbon is usually by physical adsorption and is mainly dependent on two parameters, the specific surface area and the pore size distribution. In the case of the two CNTs, the SWCNTs have a significantly larger surface area of 700 m²/g but

have a slightly lower maximum fluoride adsorption capacity of 2.4 mg/g. The pore entry diameter limits for the SWCNTs may account for the relatively low fluoride adsorption capacity of this 700 m²/g material.

Fig. 4C shows the plot of $\ln q_e$ against ε^2 , which was almost linear with a correlation coefficient (R^2) of 0.864 and 0.902 for MWCNTs and SWCNTs respectively. The mean free energy evaluated using the D–R model is 0.447 kJ mol⁻¹ for MWCNTs and SWCNTs (Table 2).

3.5. Kinetics studies

Various kinetics models (namely, pseudo-first-order (Eq. 10), pseudo-second-order (Eq. 12) and intra-particle diffusion model) were utilized to evaluate the dynamics of adsorption process [44].

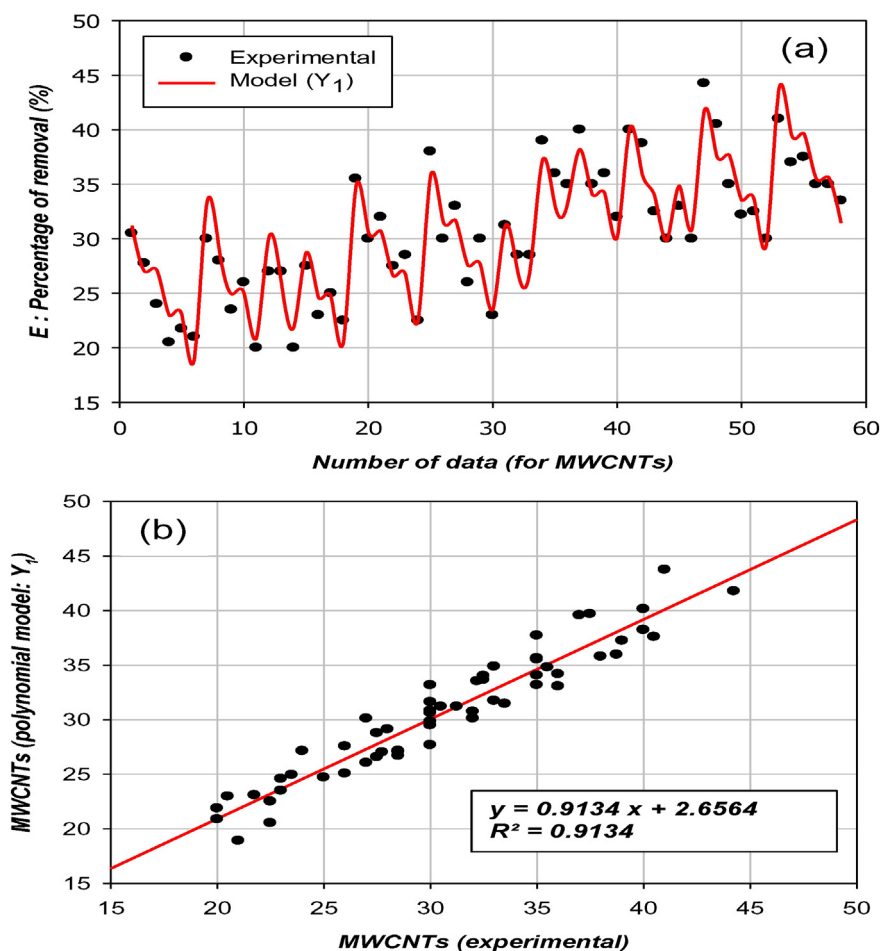


Fig. 6. (a) Agreement between the experimental data and the model outputs for MWCNTs ($R^2 = 0.913$), and (b) relationship between the proposed first-order polynomial equation (Y_1) and the experimental data obtained for MWCNTs.